

Project Title:

Dog Ownership, Infrastructure Access, and Dog Bite Incidents Across NYC Neighbourhoods (2016–2023)

Introduction

Dog ownership is common in large cities, yet the infrastructure supporting it—such as off-leash parks and dog runs—is not always evenly distributed. In New York City, open government datasets reveal clear geographic variation in both registered dog ownership and reported dog bite incidents¹. This project explores spatial patterns of dog ownership, dog-friendly infrastructure, and bite incidents across NYC to identify service gaps and public safety implications, with findings relevant to urban planners, animal welfare organisations, and public health officials.

Motivation

Having grown up in China, where dogs are banned from most public parks due to safety concerns, I was surprised by how integrated dogs are in Western cities. Yet even in New York, access to dog runs varies widely between neighbourhoods. This raised a question: if dog-friendly spaces are unevenly distributed, could this influence bite incident patterns? Exploring this may reveal whether urban infrastructure shapes safer coexistence between residents and pets.

Questions

1. How has dog ownership density varied across NYC boroughs between 2016 and 2023, and what spatial patterns emerged during the COVID-19 pandemic?
2. Are dog-friendly off-leash spaces equitably distributed relative to dog ownership density across NYC neighbourhoods, and which areas show the largest infrastructure gaps?
3. Do neighbourhoods with higher dog density and fewer off-leash spaces exhibit higher dog bite rates, and how does this pattern relate to neighbourhood income levels?

Data sources

A. **NYC Dog Licensing Dataset** – Open government dataset from NYC Health. Tabular **CSV** data with approximately **616,000 records and 17 attributes**, including breed, gender, zipcode, licence issue date, and expiry date. The dataset provides annual snapshots between **2016 and 2023**. It will be used to calculate spatial patterns of dog ownership density across neighbourhoods and identify temporal changes, such as the COVID-19 pet adoption surge. (updated on Feb 26, 2025)

Source: <https://data.cityofnewyork.us/Health/NYC-Dog-Licensing-Dataset/nu7n-tubp>

B. **DOHMH Dog Bite Data** – Incident-level public health dataset maintained by the NYC Department of Health. Tabular CSV data with about **30,000 records × 9 columns**, containing breed, borough, gender, spay/neuter status, and the date of bite incidents from **2015–2023**. This dataset will be used to analyse spatial and temporal patterns of dog bite incidents. (updated on Aug 19, 2025)

Source: <https://data.cityofnewyork.us/Health/DOHMH-Dog-Bite-Data/rsgh-akpg>

C. **NYC Dog Runs and Off-Leash Areas** – Dataset from NYC Parks describing official dog-friendly spaces. Contains **91 spatial records** including park name, borough, facility type and polygon geometry (WGS84). Although relatively small, it provides the spatial locations of dog infrastructure needed to compare with dog population distribution. (Updated on: February 22, 2026)

Source: <https://catalog.data.gov/dataset/dogruns-20190417>

D. **ACS Median Household Income by ZCTA** – Socio-economic data accessed via the R `tidycensus` package using the American Community Survey (ACS). **Tabular data** containing median household income and total population for NYC zip code areas (ZCTA) using 2018–2022 five-year estimates. This dataset will allow examination of whether dog bite patterns correlate with neighbourhood income levels.

Source: US Census Bureau via R tidycensus package

E. **NYC Zipcode Boundaries (ZCTA)** – Spatial boundary data accessed programmatically through the R “tigris” package. Contains approximately **223 polygon geometries** representing zipcode areas in NYC. This dataset will act as the spatial framework for mapping and aggregating dog ownership, infrastructure locations, and bite incidents.

Source: US Census Bureau via R tigris package (ZCTAs 2024)

¹ DogsBite.org (2025). Dog Bite Incidents in New York City: Pre- and Post-COVID Trends (2015–2023). <https://blog.dogsbite.org/2025/12/dog-bite-incidents-new-york-city-pre-post-covid-2015-2023.html>